

Vinod Kumar

8629022634 | vk5759981@gmail.com | <https://www.linkedin.com/in/vinod-kumar-a3a171370> | <https://github.com/VinodKumar836>

PROFESSIONAL SUMMARY

Data-driven CSE graduate specializing in machine learning, proficient in Python, Java, SQL and core ML concepts with hands-on experience building dedicated predictive analytical projects. Highly disciplined professional with strong analytical skills developed during intensive full time civil services preparation, now seeking a foundational role as a fresher ML Engineer.

TECHNICAL SKILLS

Languages | Python, Java, SQL, HTML/CSS (Basic)

Libraries | pandas, NumPy, Scikit-Learn, Matplotlib, Seaborn

Developer Tools | Git, GitHub, VS Code, Jupyter Notebook, MySQL

EDUCATION

B.Tech in Computer Science and Engineering | ABVGIET PRAGATINAGAR | 2024

- CGPA: 8.21/10
- Relevant Coursework: Data Structure & Algorithms, Object-Oriented Programming, Database Management, Computer Networks, Theory of Computation, Applied Statistics

Full-Time Academic Preparation (UPSC) | Independent Study | June 2024-May 2026

- Dedicated 24 months to intensive study of advanced analytical reasoning, quantitative aptitude and data interpretation under strict timelines.
- Maintained technical continuity by independently engineering standalone machine learning projects and practicing clean coding conventions in Python.

PROJECTS

House Price Prediction System | Python, Linear Regression, Scikit-Learn, Pandas, NumPy, Matplotlib

- Developed an end-to-end machine learning regression pipeline to forecast residential property values based on key regional features.
- Engineered clean data pre-processing pipelines using Pandas to handle missing data values, scale numerical features, and apply one-hot encoding to categorical variables
- Evaluated model performance using Root Mean Squared Error (RMSE) and R-squared metrics, achieving a final predictive accuracy score of [96.8%].

Breast Cancer Prediction System | Python, Support Vector Classifier, Scikit-Learn, Pandas, NumPy, Matplotlib

- Developed a binary classification machine learning model to accurately predict and diagnose the nature of breast mass tumors as malignant or benign
- Implemented a Support Vector Classifier (SVC) to map clinical features into high-dimensional space and establish an optimal separating hyperplane.
- Evaluated classification performance using a confusion matrix, achieving a score [94%] Recall rate to minimize critical false-negative predictions.

Content Recommendation System | Python, KNN-Classifer, Pandas, NumPy, Streamlit, Matplotlib

- Built a high-performance music recommendation engine utilizing a custom K-Nearest Neighbours (KNN) algorithm optimized with a KD-Tree index.
- Engineered an end-to-end production data pipeline that standardizes continuous audio vectors with StandardScaler to ensure accurate distance-space lookups.
- Developed an interactive Streamlit dashboard enabling real-time user trait manipulation